

TABEL I. DOCUMENT-GROUNDED TAXONOMY FOR URBAN PLANNING POLICY DISCOURSE

No.	Aspect Category	Operational Definition	Document-Based Scope	Examples of Terms or Expressions
1	Land Use and Land Management	Opinions addressing planning, utilization, control, compliance, permitting, or conflicts related to space and land.	RTRW: spatial patterns, spatial structure, protected areas, agricultural areas, strategic areas, and control of land use. RPJMD: implementation of spatial planning, management of location permits, land use planning, land administration, and resolution of land disputes.	spatial planning, RTRW, zoning, land, site permits, land-use conversion, zones, illegal structures, land disputes
2	Urban Infrastructure and Mobility	Opinions regarding the availability, quality, maintenance, safety, accessibility, and connectivity of infrastructure and urban transportation.	RTRW: road networks, transportation systems, urban infrastructure, and transportation zones. RPJMD: management of roads, bridges, irrigation, buildings, street lighting, traffic, road transportation, terminals, parking, and regional connectivity.	damaged roads, potholes, asphalt, bridges, sidewalks, streetlights, traffic jams, public minibuses, parking, terminals, traffic signs
3	Housing and Basic Services	Opinion pieces discussing housing conditions, the quality of residential areas, drinking water, sanitation, wastewater, neighborhood drainage, and waste management.	RTRW: areas designated for housing and settlements, water resource areas, green open spaces, and urban service facilities. RPJMD: decent housing, slum upgrading, provision of public infrastructure, facilities, and utilities, drinking water supply systems, sanitation, domestic wastewater, drainage, and solid waste management.	decent housing, RTLH, housing, slum settlements, clean water, PDAM, sanitation, septic tanks, sewers, waste, TPS, TPA
4	Environment, Climate, and Disaster Management	Opinions related to environmental quality, pollution, green open spaces, climate change, disaster risks, preparedness, and emergency response.	RTRW: water catchment areas, coastal and river buffer zones, conservation areas, areas prone to earthquakes, tsunamis, coastal erosion, and flooding, and green open spaces. RPJMD: pollution control, biodiversity management, hazardous waste management, protection of green open spaces, fire prevention, mitigation, preparedness, emergency response, and post-disaster rehabilitation.	environment, pollution, green open spaces, rivers, flooding, coastal erosion, earthquakes, tsunamis, landslides, fires, mitigation, BPBD
5	Local Economy and Livelihoods	Opinion pieces discussing employment opportunities, local businesses, trade, investment, food production, agriculture, fisheries, tourism, and affordability.	RTRW: agriculture, fisheries, industry, tourism, trade and services, mining and energy, and the informal sector. RPJMD: MSMEs, cooperatives, employment, investment, industry, trade, food security, agriculture, livestock, fisheries, tourism, the creative economy, and price stabilization.	employment, unemployment, MSMEs, businesses, capital, markets, prices, investment, merchants, farmers, fishermen, fish, tourism
6	Human Development and Social Welfare	Opinion pieces discussing education, health, poverty, social assistance, protection of vulnerable groups, and access to basic human services.	RPJMD: education, teaching staff, school facilities, health services, hospitals, community health centers, stunting, health insurance, social protection, social rehabilitation, social empowerment, services for people with disabilities, the elderly, women, and children.	schools, teachers, scholarships, education, community health centers, hospitals, BPJS, stunting, social assistance, people with disabilities, the elderly, children
7	Governance and Public Services	Opinions related to government performance, the quality of administrative services, licensing, accountability, transparency, digitization, and government data management.	RPJMD: bureaucratic reform, performance accountability, financial and asset management, local revenue, civil registration, licensing, SPBE, public communication, sectoral statistics, cryptography, records management, research, local innovation, and civil servant professionalism.	services, city government, OPDs, Bappeda, ID cards, family cards, certificates, permits, applications, SPBE, data, transparency, bureaucracy
8	Social, Cultural, and Public Order	Opinion pieces discussing security, public order, law enforcement, citizen participation, social life, culture, religion, and local identity.	RTRW: cultural heritage areas, historic city areas, places of worship, public spaces, and green open spaces. RPJMD: public peace and order, community protection, enforcement of local regulations, strengthening of the Pancasila ideology, political education, management of social conflicts, cultural development, traditional arts, cultural heritage, local history, and interfaith harmony.	safety, crime, Satpol PP, local regulations, neighborhood watch posts, Musrenbang, public input, Tabot, customs, culture, cultural heritage sites, places of worship

TABEL II. SUMMARY OF PRACTITIONER VALIDATION RESULTS

NO	EVALUATION CRITERIA	ASSESSMENT STATEMENT	SCORE	COMMENTS
1	RELEVANCE OF ASPECT CATEGORIES	The eight proposed aspect categories are relevant to the development issues that need to be considered in the planning and evaluation of Bengkulu City's development.	5	The established categories are aligned with the scope of regional development.
2	CLARITY OF CATEGORY DEFINITIONS	The operational definitions of each category are clear and understandable, allowing public opinions to be differentiated based on the development issues under discussion.	5	The definitions of each category are sufficiently clear.
3	ALIGNMENT OF TERMINOLOGY WITH BAPPEDA PRACTICES	The category names, definitions, and examples of terms are consistent with the terminology used in regional development planning documents and practices.	5	The terms used are consistent with those in the RTRW and RPJMD.
4	COMPREHENSIVENESS OF ISSUE COVERAGE	The eight categories cover key issues relevant to Bengkulu City's urban planning policies.	5	The categories cover issues that commonly arise in development planning and evaluation.
5	EXCLUSIVITY OF CATEGORIES	Each category has sufficiently clear boundaries, so that an issue does not easily overlap with another category.	4	The category boundaries are sufficiently clear, although some issues have cross-sectoral connections.
6	ADEQUACY OF THE TAXONOMY	Public opinions, complaints, or aspirations regarding the development of Bengkulu City can be grouped into the available categories.	5	The taxonomy is considered sufficient for categorizing public opinions.
7	CONCISENESS OF THE STRUCTURE	The eight categories are sufficiently concise and practical to use without omitting important development issues.	5	The eight categories are considered concise enough for the initial grouping of public opinions.
8	EASE OF EXPANSION	The taxonomy structure can be expanded by adding sub-aspects or new terms without changing the main categories.	5	The taxonomy structure is open to further expansion.
9	APPLICABILITY TO ABSA	The taxonomy can be used as a framework for grouping public opinions and for annotating or classifying aspects in ABSA for future research.	5	The taxonomy can serve as a basis for grouping public aspirations and analyzing emerging issues.

Grading Scale Explanation : 1 = strongly disagree, 2 = disagree, 3 = somewhat agree, 4 = agree, and 5 = strongly agree.

TABEL III. FINAL TAXONOMY AS AN ASPECT SCHEMA FOR ABSA

No.	Final Category	Definition	Included Issues or Sub-aspects	Category Boundaries or Exclusion Rules	Use in Advanced ABSA
1	Land Use and Land Management	A category for opinions addressing planning, utilization, control, legality, and conflicts related to space or land.	Regional Spatial Plans (RTRW), zoning, land-use designations, land-use conversion, site permits, land use, strategic areas, buildings not in compliance with their designated use, land disputes, and land management.	Complaints regarding road damage, drainage, or housing are not included in this category, unless the opinion's primary focus addresses the appropriateness of land use or land-related issues.	Used as a thematic label to identify and categorize public opinion on land use and land management.
2	Urban Infrastructure and Mobility	A category for opinions discussing the quality, availability, safety, accessibility, and maintenance of physical infrastructure and urban transportation services.	Roads, bridges, sidewalks, street lighting, irrigation, traffic congestion, public transportation, terminals, parking, traffic signs, road safety, and connectivity.	Complaints focusing on drainage, sanitation, clean water, or waste management are included in the Housing and Basic Services category.	Used as an aspect label to detect opinions related to infrastructure and mobility.
3	Housing and Basic Services	A category for opinion pieces discussing housing suitability, settlement conditions, drinking water, sanitation, wastewater, environmental drainage, and waste management.	Housing, substandard housing (RTLH), residential areas, slums, public utilities (PSU), clean water, municipal water utilities (PDAM), sanitation, septic tanks, wastewater, drainage, waste, temporary waste collection sites (TPS), and landfills (TPA).	Widespread environmental pollution, open green spaces (RTH), disaster risks, and conservation are included in the Environment, Climate, and Disasters category if they do not specifically address settlement services.	Used as aspect labels to group opinions on housing quality and basic urban services.
4	Environment, Climate, and Disasters	A category for opinion pieces discussing environmental protection, pollution, green open spaces, climate impacts, disaster risks, preparedness, and emergency response.	Water and air quality, rivers, green open spaces, pollution, hazardous waste, flooding, coastal erosion, earthquakes, tsunamis, landslides, fires, mitigation, evacuation, and disaster preparedness.	Complaints regarding household waste, drainage systems, or clean water supply are categorized under "Housing and Basic Services" if the primary focus of the opinion is the quality of basic services.	Used as an aspect label to identify opinions regarding environmental quality and disaster resilience.
5	Local Economy and Livelihoods	A category for opinion pieces discussing economic activity, employment opportunities, local businesses, trade, prices, investment, agriculture, fisheries, and tourism.	SMEs, cooperatives, employment, unemployment, markets, business capital, food prices, trade, investment, industry, agriculture, livestock, fisheries, fishermen, tourism, and the creative economy.	Social assistance, health care, education, or services for vulnerable groups are included in the Human Development and Social Welfare category if the opinion piece does not focus on economic activities or livelihoods.	Used as aspect labels to categorize opinions regarding the regional economy and people's livelihoods.
6	Human Development and Social Welfare	Categories for opinion pieces discussing education, health, poverty, social assistance, and the protection of vulnerable groups.	Schools, teachers, scholarships, education, hospitals, community health centers, BPJS, stunting, poverty, social assistance, people with disabilities, the elderly, women, children, and social rehabilitation.	Complaints regarding procedures for ID cards, family cards, certificates, permits, or government administrative responses are placed in the Governance and Public Services category.	Used as aspect labels to detect opinions regarding social services and the quality of human development.
7	Governance and Public Services	A category for opinions discussing government administration, administrative services, permits, transparency, accountability, digital services, and government officials' responsiveness.	Performance of regional government agencies (OPD), the Regional Development Planning Agency (Bappeda), public services, ID cards (KTP), family cards (KK), certificates, permits, government applications, the Electronic Government System (SPBE), data, transparency, budgeting, bureaucracy, and service complaints.	Issues related to crime, public order, social conflict, enforcement of local regulations, and environmental safety are included in the category of Social, Cultural, and Public Order Life.	These are used as aspect labels to group opinions regarding the quality of government administration and public services.
8	Social, Cultural, and Public Order	A category for opinions discussing security, public order, legal compliance, citizen participation, local culture, religion, and the social identity of the community.	Crime, Satpol PP, local regulations, neighborhood watch posts, community safety, Musrenbang, residents' concerns, social conflicts, Tabot, customs, the arts, cultural heritage sites, local history, places of worship, and community harmony.	Complaints primarily concerning population administration, licensing, digital services, or bureaucratic processes are categorized under Governance and Public Services.	Used as aspect labels to detect opinions regarding social, cultural, security, and public order conditions.